IMMERSIVE 3D MAP VIEWS @MAPLIBRE

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Image: AI-generated

CONTEXT

- Enable efficient cartography-oriented 3D visualization within the MapLibre ecosystem
- Support low-latency rendering while keeping the familiar “slippy map” experience on mobile phones and low-bandwidth networks
- Focus on moderate Levels of Detail sufficient for orientation, navigation and spatial understanding → mainly LOD1–LOD2 buildings, with selected LOD3 landmarks
- Inspired by Google Maps’ Immersive Navigation and Apple Maps’ Detailed City Experience
- Generate 3D map experiences from open data such as OpenStreetMap and Overture Maps



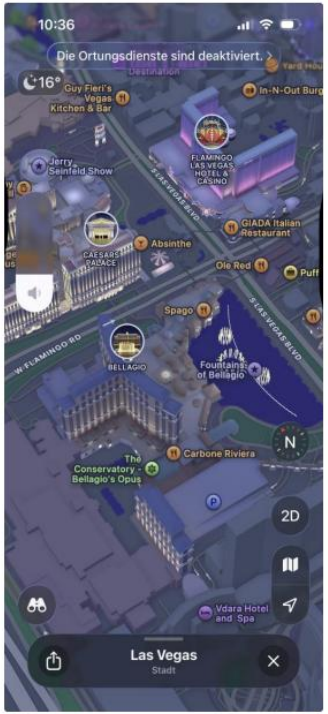
Source: Google Maps Platform Blog



Source: Apple Maps, iDROPNEWS

Goals: Cartography-Oriented 3D Visualization (Symbolised 3D Models)

Examples: Google Maps Immersive Navigation, Apple Maps Detailed City Experience



Source: Apple Maps screenshot © Apple Inc.

MLT 3D

Non-Goals: Photorealistic Rendering

(not a primary focus, as this is already well supported by 3D Tiles)

Examples: Google Maps Photorealistic 3D Tiles



Source: Apple Maps screenshot © Apple Inc.

3DTiles: massive 3D geodata content such as photogrammetry models, BIM/CAD, digital twins, point clouds, gaussian splatting

LODs

Primary focus on navigation, enabling large-scale generation from open data sources like OSM and Overture Maps

LOD 1



MASS MODEL

FOCUS ON SHAPE, SIZE & LOCATION

LOD 2



SIMILAR TO 1:100 SURVEY

FOCUS ON FOUNDATIONAL ELEMENTS

LOD 3



SIMILAR TO 1:50 SURVEY

PRECISE STRUCTURE AND WINDOWS/DOOR SIZING

LOD 4



SIMILAR TO 1:20 SURVEY

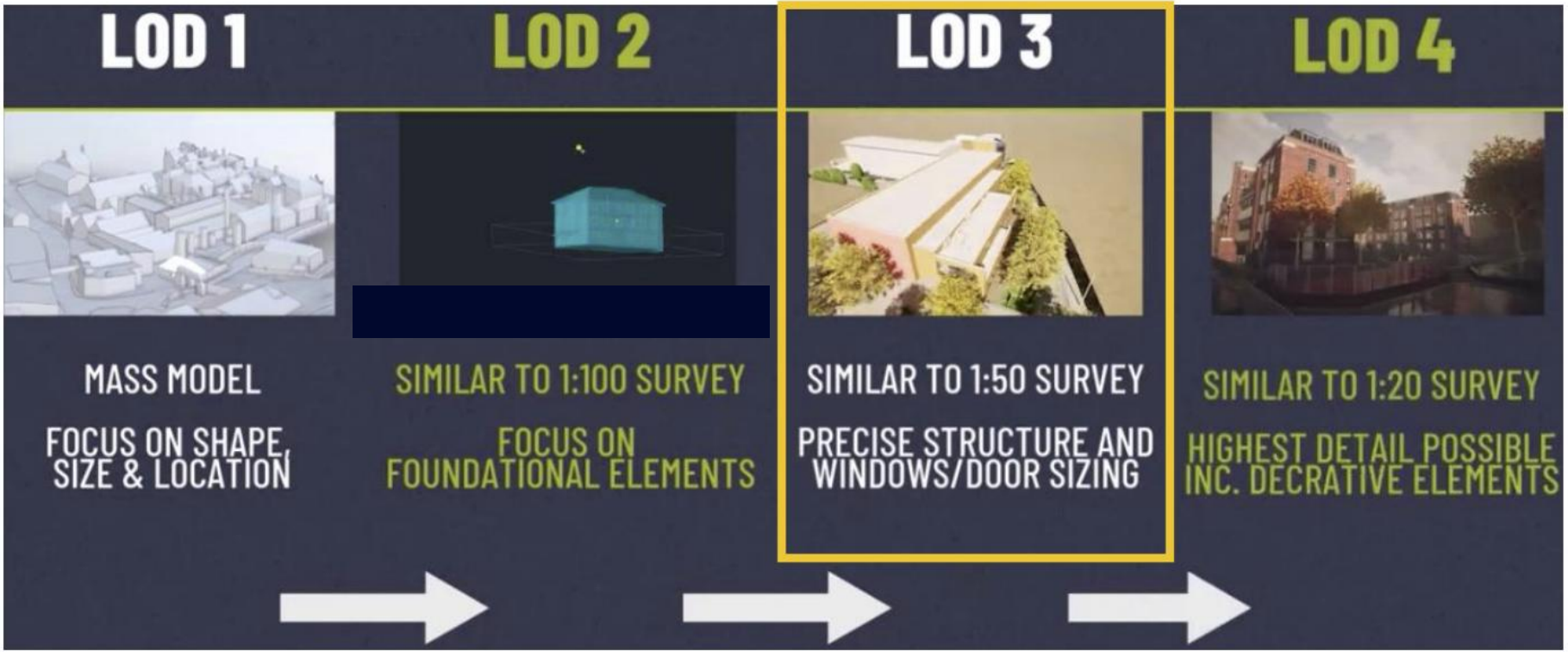
HIGHEST DETAIL POSSIBLE INC. DECRATIVE ELEMENTS

Source: Spatial Dimensions

LODs

Can include key landmarks and dominant buildings, but not optimized for streaming a large number of LOD3 models in a single view

→ First versions: LOD selection via tile zoom level only, no advanced LOD selection optimized for massive 3D datasets (e.g., geometric error / screen-space error as in 3D Tiles)



Source: Spatial Dimensions

IMPLEMENTATION

Goals

- **Stylable**: integrated into the MapLibre ecosystem
- **Efficient**: fast to process and render on the web, on mobile devices and over constrained networks
- **Visually plausible, not photorealistic**: symbolic realism reduces cognitive load

Components

- **Renderer extension (GL JS and Native):** MLT 3D support plus selected techniques for lighting, shading, shadows and anti-aliasing
- **MLT 3D format**: efficient format optimized for 3D cartographic visualization
- **Data generation pipeline**: extended generators that create MLT 3D content from open geospatial data
- **Style specification**: extended styling rules for 3D cartographic visualization

MAPLIBRE 3D: FOUR TIERS

- Tier 0 (LOD1) → MLT, MVT
- Tier 1 (constrained LOD2) → MLT 3D or MLT/MVT
- Tier 2 (full LOD2 / LOD3) → MLT 3D, glTF or 3D Tiles
- Tier 3 (LOD3+) → 3D Tiles

MAPLIBRE 3D: FOUR TIERS

- **Tier 0 (LOD1) → MLT, MVT**
  - Already supported: building footprints with height attributes (e.g., render_height, render_min_height), drawn with a fill-extrusion layer
- Tier 1 (constrained LOD2) → MLT 3D or MLT/MVT?
- Tier 2 (full LOD2 / LOD3) → MLT 3D, glTF or 3D Tiles?
- Tier 3 (LOD3+) → 3D Tiles

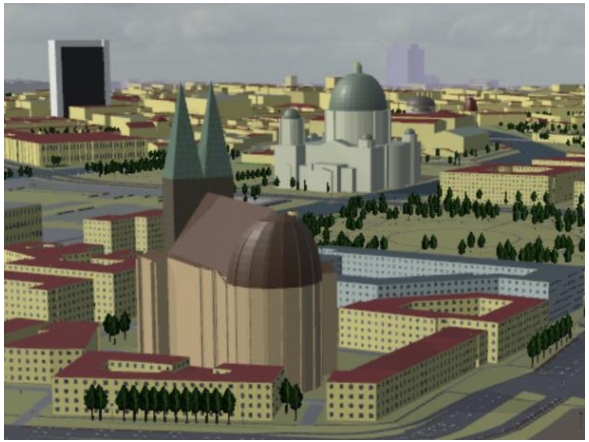


MAPLIBRE 3D: FOUR TIERS

- Tier 0 (LOD1) → MLT, MVT
- **Tier 1 (constrained LOD2) → MLT 3D or MLT/MVT?**
  - Adds support for common roof types and 3D extruded roads with overpasses
  - Stores parametric and procedural 2.5D/3D objects as compact geometries rather than pre-tessellated meshes
  - Either as a dedicated geometry type in MLT (e.g., ExtrusionObject), or as feature attributes in MLT/MVT
  - Core paradigm: **compact representation → GPU expansion** (e.g., via WebGPU compute shaders or WebGL 2 vertex texture fetch) → **rendering**
- Tier 2 (full LOD2 / LOD3) → MLT 3D, glTF or 3D Tiles?
- Tier 3 (LOD3+) → 3D Tiles



Source: Google Maps Immersive Navigation, Pocket-lint



Source: Osm2World

MAPLIBRE 3D: FOUR TIERS

- Tier 0 (LOD1) → MLT, MVT
- Tier 1 (constrained LOD2) → MLT 3D or MLT/MVT?
- **Tier 2 (full LOD2 / LOD3) → MLT 3D, glTF or 3D Tiles?**
  - Adds support for landmark buildings and shapes that can't be parametrically described
  - Options
    - **MLT 3D GPUTables** (e.g., SceneTable): flat, columnar mesh storage, clearly simplified compared to glTF and native MapLibre styling
    - **glTF** embedded in MLT: reuses the glTF ecosystem and mesh compression
    - **3D Tiles** as separate datasets: loaded alongside the MLT tiles
- Tier 3 (LOD3+) → 3D Tiles



Source: Apple Maps, iDROPNEWS

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- **Tier 3 (LOD3+) → 3D Tiles**
  - Requires advanced 3D format concepts: 3D spatial data structures (e.g., octrees), hierarchical LOD (HLOD) with geometric-error-based selection, refinement strategies, …
  - Typically paired with photorealistic rendering: PBR materials, HDR lighting with tone mapping, and post-processing effects such as bloom, lens flare and depth of field



Source: Cesium Sandcastle



Source: Google Maps Platform

VISUAL 3D COMPUTER GRAPHICS CONCEPTS

Existing: one directional light, Lambert diffuse shading calculated per vertex (Gouraud), vertical gradient on fill extrusions

Candidate features for MapLibre GL JS and MapLibre Native

- Shading and materials: emissive surfaces, PBR with the metallic-roughness model, …
- Ambient occlusion: analytic (maplibre-building-shadows) or screen-space (SSAO)
- Shadows: projected footprint shadows (planar, maplibre-building-shadows), shadow mapping with cascades
- Anti-aliasing: MSAA, FXAA or TAA
- Post-processing and screen-space effects: tone mapping, bloom, depth of field, lens flare, screen-space reflections (SSR)
- …

MVP1 Proposal: cascaded shadows, ambient occlusion, and (proper) vertical building gradients

LET'S DISCUSS

MAPLIBRE IMMERSIVE 3D WORKING GROUP

Interested in joining?

Coordinate format, renderer, style specification and data generation together

– Shared roadmap and priorities
– Regular calls
– Aligned changes across GL JS, Native and the style spec

→ Join #maplibre-3d on Slack and the monthly Immersive 3D Maps Working Group meeting; 3D topics are also covered in the monthly MLT meeting

YOUR OPINIONS

1 Does the tier model (MLT 3D vs. 3D Tiles) fit your use cases?

2 Which rendering techniques matter most to you for MVP1?

3 What do you need from the style specification for 3D?

4 Where could you contribute: data, format, GL JS or Native?